

Homework 33: Least Squares Problem II

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BCS A, E, F, G

Problem 1. Infinitely Many Least Squares Solutions. Find the least squares solution, the least squares error vector, and the least squares error of the linear system

$$3x_1 + 2x_2 - x_3 = 2, \quad x_1 - 4x_2 + 3x_3 = -2, \quad x_1 + 10x_2 - 7x_3 = 1.$$

Problem 2. Continuation of The Moore-Penrose Pseudoinverse.

Given a matrix

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}.$$

- a. Find the left inverse of A .
- b. Find the right inverse of A .

In the following Theorem, we will discuss what can be done in the cases when $A^T A$ or AA^T are not invertible.

Theorem. Tikhonov's Regularization.

For all $A \in \mathbb{R}_r^{m \times n}$ one has

$$\begin{aligned} A^+ &= \lim_{\delta \rightarrow 0} A^T (AA^T + \delta^2 I)^{-1} \\ &= \lim_{\delta \rightarrow 0} (A^T A + \delta^2 I)^{-1} A^T. \end{aligned}$$

Here $\mathbb{R}_r^{m \times n}$ the set of real $m \times n$ matrices of rank r . Thus, $\mathbb{R}_n^{n \times n}$ denotes the set of real nonsingular $n \times n$ matrices.

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Good Luck